

Hard-to-Cache Identification Table (HCIT)

A table indexed by instruction PC stores hit and miss counts. PCs exhibiting high miss-to-hit ratios are classified as hard-to-cache.

Difficulty Metric

Difficulty Score =
 $\text{Misses} / (\text{Hits} + 1)$

A high score indicates that accesses generated by the PC are difficult to retain in cache.

Adaptive Protection

Blocks originating from hard-to-cache PCs are inserted with:

RRPV = 0

rather than:

RRPV = 2

used by SRRIP.

This delays eviction and increases reuse opportunities.

Expected Benefits

Higher LLC hit rate
Lower LLC miss rate
Higher IPC

particularly for:

605.mcf
462.libquantum

which are memory-intensive workloads.